

Experiment No. 5

Title

Vacuum Cleaner Problem Using Artificial Intelligence

Aim

To develop a Python program that implements the Vacuum Cleaner Problem using a Simple Reflex Agent.

Objective

The objective of this experiment is to understand the working of a Simple Reflex Agent by implementing a vacuum cleaner that senses the environment and cleans dirty rooms automatically.

Algorithm

Step 1: Define the rooms and their initial states (Clean/Dirty).

Step 2: Place the vacuum cleaner in the first room.

Step 3: Check whether the current room is dirty.

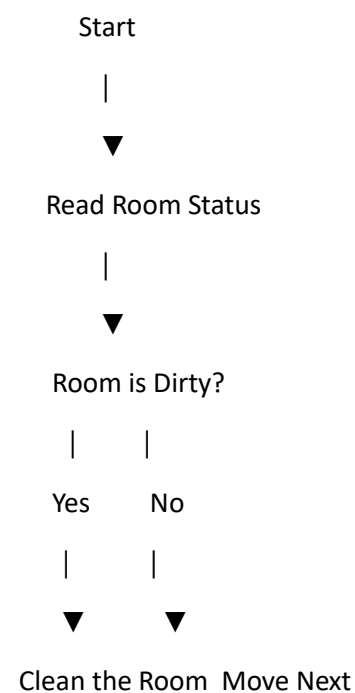
Step 4: If dirty, clean the room.

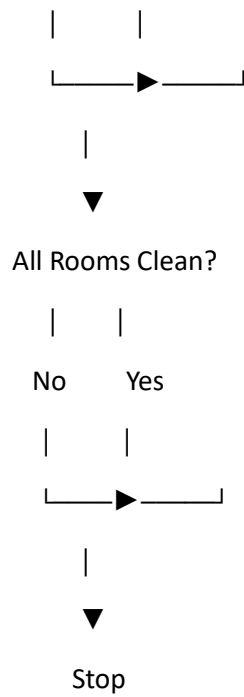
Step 5: Move to the next room.

Step 6: Repeat until all rooms are cleaned.

Step 7: Stop.

Flowchart





Python Program

```
rooms = {  
    "A": "Dirty",  
    "B": "Dirty"  
}  
  
for room in rooms:  
    print(f"Room {room} is {rooms[room]}")  
  
    if rooms[room] == "Dirty":  
        print(f"Cleaning Room {room}...")  
        rooms[room] = "Clean"  
  
print("\nFinal Room Status:")  
for room in rooms:  
    print(f"Room {room}: {rooms[room]}")
```

Sample Output

Room A is Dirty

Cleaning Room A...

Room B is Dirty

Cleaning Room B...

Final Room Status:

Room A: Clean

Room B: Clean

Result

The Vacuum Cleaner Agent successfully cleaned all the dirty rooms.

Conclusion

Thus, the Vacuum Cleaner Problem was successfully implemented using a Simple Reflex Agent, which senses the environment and cleans all dirty rooms automatically.